

Vector Calculus [Double, Triple, Multiple (Jacobeau) Integral]

1. Evaluated the iterated integrals: DOUBLE Integral

$$(a) \int_{\frac{\pi}{2}}^{\pi} \int_0^{x^2} \frac{1}{x} \cos \frac{y}{x} dy dx \quad (b) \int_0^1 \int_0^1 \frac{x}{(xy+1)^2} dy dx \quad (c) \int_1^2 \int_0^{y^2} e^{\frac{x}{y^2}} dx dy \quad (d) \int_0^2 \int_0^{\sqrt{4-x^2}} e^{\sqrt{x^2+y^2}} dy dx.$$

2. DOUBLE Integral

(a) Find the area of the region inside the circle $r = 4\sin\theta$ and outside the circle $r = 2$.

(b) $\iint_R \frac{1}{x^2 + y^2 + 1} dA$, where R is the sector in the first quadrant bounded by
 $y = 0$, $y = x$ and $x^2 + y^2 = 4$.

3. Use polar coordinates to evaluate the double integral $\int_{-a}^a \int_0^{\sqrt{a^2-x^2}} (x^2 + y^2)^{1/2} dy dx$. (DOUBLE Int.)

4. (a) Find the volume of the solid that is bounded by the cylinder $y = x^2$ and by the planes $y + z = 4$ and $z = 0$. (TRIPLE Int)

(b) Find the volume of the surface enclosed by the surfaces $z = x^2 + 3y^2$ and $z = 8 - x^2 - y^2$. (TRIPLE Int)

5. Evaluated the iterated integral by converting to polar coordinates: (DOUBLE Int)

$$(a) \int_0^1 \int_0^{\sqrt{1-x^2}} (x^2 + y^2) dy dx \quad (b) \int_0^2 \int_0^{\sqrt{2x-x^2}} \sqrt{x^2 + y^2} dy dx$$

$$(c) \int_0^a \int_0^{\sqrt{a^2-x^2}} \frac{dy dx}{(1+x^2+y^2)^{\frac{3}{2}}} \quad (a > 0).$$

6. (a) Evaluate $\int_{y=0}^4 \int_{x=\frac{y}{2}}^{\frac{y}{2}+1} \frac{2x-y}{2} dx dy$ by applying transformation T :

where $u = \frac{2x-y}{2}$, $v = \frac{y}{2}$ and integrating over an appropriate region in uv -plane. (JACOBEAN)

(b) Evaluate $\iint_R \frac{x-y}{x+y} dA$, where R is the region enclosed by the lines

$x - y = 0$, $x - y = 1$, $x + y = 1$ & $x + y = 3$, using the transformation.

(JACOBEAN)

Double Integral

Exercise- 14.1- 1-16.

EXERCISE SET 14.1

 CAS

1–12 Evaluate the iterated integrals. ■

1. $\int_0^1 \int_0^2 (x+3) dy dx$
2. $\int_1^3 \int_{-1}^1 (2x-4y) dy dx$
3. $\int_2^4 \int_0^1 x^2 y dx dy$
4. $\int_{-2}^0 \int_{-1}^2 (x^2 + y^2) dx dy$
5. $\int_0^{\ln 3} \int_0^{\ln 2} e^{x+y} dy dx$
6. $\int_0^2 \int_0^1 y \sin x dy dx$
7. $\int_{-1}^0 \int_2^5 dx dy$
8. $\int_4^6 \int_{-3}^7 dy dx$
9. $\int_0^1 \int_0^1 \frac{x}{(xy+1)^2} dy dx$
10. $\int_{\pi/2}^{\pi} \int_1^2 x \cos xy dy dx$
11. $\int_0^{\ln 2} \int_0^1 xy e^{y^2 x} dy dx$
12. $\int_3^4 \int_1^2 \frac{1}{(x+y)^2} dy dx$

13–16 Evaluate the double integral over the rectangular region R . ■

13. $\iint_R 4xy^3 dA$; $R = \{(x, y) : -1 \leq x \leq 1, -2 \leq y \leq 2\}$
14. $\iint_R \frac{xy}{\sqrt{x^2 + y^2 + 1}} dA$;
 $R = \{(x, y) : 0 \leq x \leq 1, 0 \leq y \leq 1\}$
15. $\iint_R x\sqrt{1-x^2} dA$; $R = \{(x, y) : 0 \leq x \leq 1, 2 \leq y \leq 3\}$
16. $\iint_R (x \sin y - y \sin x) dA$;
 $R = \{(x, y) : 0 \leq x \leq \pi/2, 0 \leq y \leq \pi/3\}$

Exercise- 14.2- 1-26.

EXERCISE SET 14.2

Graphing Utility CAS

1–8 Evaluate the iterated integral. ■

1. $\int_0^1 \int_{x^2}^x xy^2 dy dx$
2. $\int_1^{3/2} \int_y^{3-y} y dx dy$
3. $\int_0^3 \int_0^{\sqrt{9-y^2}} y dx dy$
4. $\int_{1/4}^1 \int_{x^2}^x \sqrt{x} dy dx$
5. $\int_{\sqrt{\pi}}^{\sqrt{2\pi}} \int_0^{x^3} \sin \frac{y}{x} dy dx$
6. $\int_{-1}^1 \int_{-x^2}^{x^2} (x^2 - y) dy dx$
7. $\int_0^1 \int_0^x y\sqrt{x^2 - y^2} dy dx$
8. $\int_1^2 \int_0^{y^2} e^{x/y^2} dx dy$

FOCUS ON CONCEPTS

9. Let R be the region shown in the accompanying figure. Fill in the missing limits of integration.

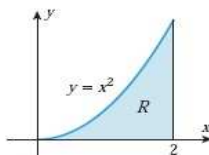
(a) $\iint_R f(x, y) dA = \int_{\square} \int_{\square} f(x, y) dy dx$

(b) $\iint_R f(x, y) dA = \int_{\square} \int_{\square} f(x, y) dx dy$

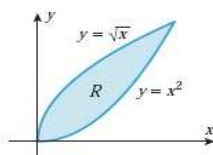
10. Let R be the region shown in the accompanying figure. Fill in the missing limits of integration.

(a) $\iint_R f(x, y) dA = \int_{\square} \int_{\square} f(x, y) dy dx$

(b) $\iint_R f(x, y) dA = \int_{\square} \int_{\square} f(x, y) dx dy$



▲ Figure Ex-9



▲ Figure Ex-10

14. Evaluate $\iint_R (x + y) dA$, where R is the region in

(a) Exercise 10

(b) Exercise 12.

15–18 Evaluate the double integral in two ways using iterated integrals: (a) viewing R as a type I region, and (b) viewing R as a type II region. ■

15. $\iint_R x^2 dA$; R is the region bounded by $y = 16/x$, $y = x$, and $x = 8$.

16. $\iint_R xy^2 dA$; R is the region enclosed by $y = 1$, $y = 2$, $x = 0$, and $y = x$.

17. $\iint_R (3x - 2y) dA$; R is the region enclosed by the circle $x^2 + y^2 = 1$.

18. $\iint_R y dA$; R is the region in the first quadrant enclosed between the circle $x^2 + y^2 = 25$ and the line $x + y = 5$.

19–24 Evaluate the double integral. ■

19. $\iint_R x(1 + y^2)^{-1/2} dA$; R is the region in the first quadrant enclosed by $y = x^2$, $y = 4$, and $x = 0$.

20. $\iint_R x \cos y dA$; R is the triangular region bounded by the lines $y = x$, $y = 0$, and $x = \pi$.

21. $\iint_R xy dA$; R is the region enclosed by $y = \sqrt{x}$, $y = 6 - x$, and $y = 0$.

11. Let R be the region shown in the accompanying figure. Fill in the missing limits of integration.

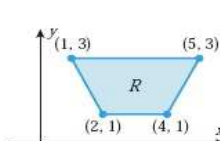
(a) $\iint_R f(x, y) dA = \int_1^2 \int_{\square} f(x, y) dy dx$
 $+ \int_2^4 \int_{\square} f(x, y) dy dx$
 $+ \int_4^5 \int_{\square} f(x, y) dy dx$

(b) $\iint_R f(x, y) dA = \int_{\square} \int_{\square} f(x, y) dx dy$

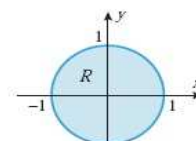
12. Let R be the region shown in the accompanying figure. Fill in the missing limits of integration.

(a) $\iint_R f(x, y) dA = \int_{\square} \int_{\square} f(x, y) dy dx$

(b) $\iint_R f(x, y) dA = \int_{\square} \int_{\square} f(x, y) dx dy$



▲ Figure Ex-11



▲ Figure Ex-12

13. Evaluate $\iint_R xy dA$, where R is the region in

(a) Exercise 9

(b) Exercise 11.

20. $\iint_R x \cos y dA$; R is the triangular region bounded by the lines $y = x$, $y = 0$, and $x = \pi$.

21. $\iint_R xy dA$; R is the region enclosed by $y = \sqrt{x}$, $y = 6 - x$, and $y = 0$.

22. $\iint_R x dA$; R is the region enclosed by $y = \sin^{-1} x$, $x = 1/\sqrt{2}$, and $y = 0$.

23. $\iint_R (x - 1) dA$; R is the region in the first quadrant enclosed between $y = x$ and $y = x^3$.

24. $\iint_R x^2 dA$; R is the region in the first quadrant enclosed by $xy = 1$, $y = x$, and $y = 2x$.

25. Evaluate $\iint_R \sin(y^3) dA$, where R is the region bounded by $y = \sqrt{x}$, $y = 2$, and $x = 0$. [Hint: Choose the order of integration carefully.]

26. Evaluate $\iint_R x dA$, where R is the region bounded by $x = \ln y$, $x = 0$, and $y = e$.

Exercise- 14.3- 1-12, 23-34.

1. $\int_0^{\pi/2} \int_0^{\sin \theta} r \cos \theta \, dr \, d\theta$
2. $\int_0^{\pi} \int_0^{1+\cos \theta} r \, dr \, d\theta$
3. $\int_0^{\pi/2} \int_0^{a \sin \theta} r^2 \, dr \, d\theta$
4. $\int_0^{\pi/6} \int_0^{\cos 3\theta} r \, dr \, d\theta$
5. $\int_0^{\pi} \int_0^{1-\sin \theta} r^2 \cos \theta \, dr \, d\theta$
6. $\int_0^{\pi/2} \int_0^{\cos \theta} r^3 \, dr \, d\theta$

7-10 Use a double integral in polar coordinates to find the area of the region described. ■

7. The region enclosed by the cardioid $r = 1 - \cos \theta$.
8. The region enclosed by the rose $r = \sin 2\theta$.
9. The region in the first quadrant bounded by $r = 1$ and $r = \sin 2\theta$, with $\pi/4 \leq \theta \leq \pi/2$.
10. The region inside the circle $x^2 + y^2 = 4$ and to the right of the line $x = 1$.

FOCUS ON CONCEPTS

11-12 Let R be the region described. Sketch the region R and fill in the missing limits of integration. ■

$$\iint_R f(r, \theta) \, dA = \int_{\square} \int_{\square} f(r, \theta) r \, dr \, d\theta \quad \blacksquare$$

11. The region inside the circle $r = 4 \sin \theta$ and outside the circle $r = 2$.
12. The region inside the circle $r = 1$ and outside the cardioid $r = 1 + \cos \theta$.

13-16 Express the volume of the solid described as a double integral in polar coordinates. ■

23. $\iint_R \sin(x^2 + y^2) \, dA$, where R is the region enclosed by the circle $x^2 + y^2 = 9$.
24. $\iint_R \sqrt{9 - x^2 - y^2} \, dA$, where R is the region in the first quadrant within the circle $x^2 + y^2 = 9$.
25. $\iint_R \frac{1}{1 + x^2 + y^2} \, dA$, where R is the sector in the first quadrant bounded by $y = 0$, $y = x$, and $x^2 + y^2 = 4$.
26. $\iint_R 2y \, dA$, where R is the region in the first quadrant bounded above by the circle $(x - 1)^2 + y^2 = 1$ and below by the line $y = x$.

27-34 Evaluate the iterated integral by converting to polar coordinates. ■

27. $\int_0^1 \int_0^{\sqrt{1-x^2}} (x^2 + y^2) \, dy \, dx$
28. $\int_{-2}^2 \int_{-\sqrt{4-y^2}}^{\sqrt{4-y^2}} e^{-(x^2+y^2)} \, dx \, dy$
29. $\int_0^2 \int_0^{\sqrt{2x-x^2}} \sqrt{x^2 + y^2} \, dy \, dx$
30. $\int_0^1 \int_0^{\sqrt{1-y^2}} \cos(x^2 + y^2) \, dx \, dy$
31. $\int_0^a \int_0^{\sqrt{a^2-x^2}} \frac{dy \, dx}{(1 + x^2 + y^2)^{3/2}} \quad (a > 0)$
32. $\int_0^1 \int_y^{\sqrt{y}} \sqrt{x^2 + y^2} \, dx \, dy$
33. $\int_0^{\sqrt{2}} \int_y^{\sqrt{4-y^2}} \frac{1}{\sqrt{1 + x^2 + y^2}} \, dx \, dy$
34. $\int_{-4}^0 \int_{-\sqrt{16-x^2}}^{\sqrt{16-x^2}} 3x \, dy \, dx$

Surface Area from Double Integral

Exercise- 14.4- 1-9.

EXERCISE SET 14.4 Graphing Utility CAS

1-4 Express the area of the given surface as an iterated double integral, and then find the surface area. ■

1. The portion of the cylinder $y^2 + z^2 = 9$ that is above the rectangle $R = \{(x, y) : 0 \leq x \leq 2, -3 \leq y \leq 3\}$.
2. The portion of the plane $2x + 2y + z = 8$ in the first octant.
3. The portion of the cone $z^2 = 4x^2 + 4y^2$ that is above the region in the first quadrant bounded by the line $y = x$ and the parabola $y = x^2$.
4. The portion of the surface $z = 2x + y^2$ that is above the triangular region with vertices $(0, 0)$, $(0, 1)$, and $(1, 1)$.

5-10 Express the area of the given surface as an iterated double integral in polar coordinates, and then find the surface area. ■

5. The portion of the cone $z = \sqrt{x^2 + y^2}$ that lies inside the cylinder $x^2 + y^2 = 2x$.
6. The portion of the paraboloid $z = 1 - x^2 - y^2$ that is above the xy -plane.
7. The portion of the surface $z = xy$ that is above the sector in the first quadrant bounded by the lines $y = x/\sqrt{3}$, $y = 0$, and the circle $x^2 + y^2 = 9$.
8. The portion of the paraboloid $2z = x^2 + y^2$ that is inside the cylinder $x^2 + y^2 = 8$.
9. The portion of the sphere $x^2 + y^2 + z^2 = 16$ between the planes $z = 1$ and $z = 2$.

Triple Integral

Exercise- 14.5- 1-12, 15-18.

EXERCISE SET 14.5 CAS

1-8 Evaluate the iterated integral. ■

- $\int_{-1}^1 \int_0^2 \int_0^1 (x^2 + y^2 + z^2) dx dy dz$
- $\int_{1/3}^{1/2} \int_0^\pi \int_0^1 zx \sin xy dz dy dx$
- $\int_0^2 \int_{-1}^{y^2} \int_{-1}^z yz dx dz dy$
- $\int_0^{\pi/4} \int_0^1 \int_0^{x^2} x \cos y dz dx dy$

9-12 Evaluate the triple integral. ■

- $\iiint_G xy \sin yz dV$, where G is the rectangular box defined by the inequalities $0 \leq x \leq \pi$, $0 \leq y \leq 1$, $0 \leq z \leq \pi/6$.
- $\iiint_G y dV$, where G is the solid enclosed by the plane $z = y$, the xy -plane, and the parabolic cylinder $y = 1 - x^2$.
- $\iiint_G xyz dV$, where G is the solid in the first octant that is bounded by the parabolic cylinder $z = 2 - x^2$ and the planes $z = 0$, $y = x$, and $y = 0$.
- $\iiint_G \cos(z/y) dV$, where G is the solid defined by the inequalities $\pi/6 \leq y \leq \pi/2$, $y \leq x \leq \pi/2$, $0 \leq z \leq xy$.

- $\int_0^3 \int_0^{\sqrt{9-z^2}} \int_0^x xy dy dx dz$
- $\int_1^3 \int_x^{x^2} \int_0^{\ln z} xe^y dy dz dx$
- $\int_0^2 \int_0^{\sqrt{4-x^2}} \int_{-5+x^2+y^2}^{3-x^2-y^2} x dz dy dx$
- $\int_1^2 \int_z^2 \int_0^{\sqrt{3}y} \frac{y}{x^2 + y^2} dx dy dz$

15-18 Use a triple integral to find the volume of the solid. ■

- The solid in the first octant bounded by the coordinate planes and the plane $3x + 6y + 4z = 12$.
- The solid bounded by the surface $z = \sqrt{y}$ and the planes $x + y = 1$, $x = 0$, and $z = 0$.
- The solid bounded by the surface $y = x^2$ and the planes $y + z = 4$ and $z = 0$.
- The wedge in the first octant that is cut from the solid cylinder $y^2 + z^2 \leq 1$ by the planes $y = x$ and $x = 0$.

Change of variables

Exercise- 14.7- 1-12, 21-24, 35-37.

EXERCISE SET 14.7

1-4 Find the Jacobian $\partial(x, y)/\partial(u, v)$. ■

- $x = u + 4v$, $y = 3u - 5v$
- $x = u + 2v^2$, $y = 2u^2 - v$
- $x = \sin u + \cos v$, $y = -\cos u + \sin v$
- $x = \frac{2u}{u^2 + v^2}$, $y = -\frac{2v}{u^2 + v^2}$

5-8 Solve for x and y in terms of u and v , and then find the Jacobian $\partial(x, y)/\partial(u, v)$. ■

- $u = 2x - 5y$, $v = x + 2y$
- $u = e^x$, $v = ye^{-x}$
- $u = x^2 - y^2$, $v = x^2 + y^2$ ($x > 0, y > 0$)
- $u = xy$, $v = xy^3$ ($x > 0, y > 0$)

9-12 Find the Jacobian $\partial(x, y, z)/\partial(u, v, w)$. ■

- $x = 3u + v$, $y = u - 2w$, $z = v + w$
- $x = u - uv$, $y = uv - uvw$, $z = uvw$
- $u = xy$, $v = y$, $w = x + z$
- $u = x + y + z$, $v = x + y - z$, $w = x - y + z$

21. Use the transformation $u = x - 2y$, $v = 2x + y$ to find

$$\iint_R \frac{x - 2y}{2x + y} dA \quad (\text{com.})$$

where R is the rectangular region enclosed by the lines $x - 2y = 1$, $x - 2y = 4$, $2x + y = 1$, $2x + y = 3$.

22. Use the transformation $u = x + y$, $v = x - y$ to find

$$\iint_R (x - y)e^{x^2 - y^2} dA$$

over the rectangular region R enclosed by the lines $x + y = 0$, $x + y = 1$, $x - y = 1$, $x - y = 4$.

23. Use the transformation $u = \frac{1}{2}(x + y)$, $v = \frac{1}{2}(x - y)$ to find

$$\iint_R \sin \frac{1}{2}(x + y) \cos \frac{1}{2}(x - y) dA$$

over the triangular region R with vertices $(0, 0)$, $(2, 0)$, $(1, 1)$.

24. Use the transformation $u = y/x$, $v = xy$ to find

$$\iint_R xy^3 dA$$

over the region R in the first quadrant enclosed by $y = x$, $y = 3x$, $xy = 1$, $xy = 4$.

35–38 Evaluate the integral by making an appropriate change of variables. ■

35. $\iint_R \frac{y - 4x}{y + 4x} dA$, where R is the region enclosed by the lines $y = 4x$, $y = 4x + 2$, $y = 2 - 4x$, $y = 5 - 4x$.

36. $\iint_R (x^2 - y^2) dA$, where R is the rectangular region enclosed by the lines $y = -x$, $y = 1 - x$, $y = x$, $y = x + 2$.

37. $\iint_R \frac{\sin(x - y)}{\cos(x + y)} dA$, where R is the triangular region enclosed by the lines $y = 0$, $y = x$, $x + y = \pi/4$.

Book: Elementary Calculus- Howard Anton (10th Edition), Soft Copy